

Quiz (Carolina Reaper)

- Q1.** A shipping company increased its delivery rate by 15%. If the original rate was 240 packages per day, what is the new rate?
(A) $240 + 15$
(B) $240 + 24$
(C) $240 + 36$
(D) 276
(E) 300
- Q2.** A traffic congestion reduced the average speed of a highway by 12%. If the original speed was 60 miles per hour, what is the reduced speed?
(A) 48
(B) 50
(C) 52
(D) 53
(E) 55
- Q3.** The cost of fuel for a logistic company increased from 2.50 to 3.15 per gallon. What is the percent increase?
(A) 20%
(B) 25%
(C) 26%
(D) 30%
(E) 35%
- Q4.** A transportation company has 480 buses. If 15% of the buses are currently under maintenance, how many buses are operational?
(A) 384
(B) 408
(C) 416
(D) 424
- (E) 432
- Q5.** A new traffic management system reduced congestion by 8% on a major highway. If the original number of congested hours was 48 per day, what is the new number of congested hours?
(A) 36
(B) 40
(C) 41.6
(D) 44
(E) 46
- Q6.** A delivery service promises to deliver 95% of its packages within two days. If the service handles 1000 packages per day, how many packages will be delivered within two days?
(A) 900
(B) 925
(C) 950
(D) 975
(E) 1000
- Q7.** A transportation company reduced its carbon footprint by 22%. If the original footprint was 550 tons per year, what is the new carbon footprint?
(A) 350
(B) 381
(C) 415
(D) 429
(E) 450
- Q8.** A shipping container can hold 2500 packages. If it is currently 80% full, how many more packages can be added?
(A) 200
(B) 250
(C) 300
(D) 400
(E) 500

Open Ended Questions (FRQ)

- Q9.** A traffic flow study found that 62% of the cars on the road during rush hour were single-occupant vehicles. If there were 1500 cars on the road during rush hour, how many were single-occupant vehicles? Show your work and calculate the number of single-occupant vehicles.
- Q10.** A logistics company wants to reduce its delivery time by 18%. If the current average delivery time is 36 hours, what should be the new average delivery time? Explain your reasoning and show your calculations.

Chef's Tips

- Emphasize the importance of reading questions carefully and understanding the context of the problem.
- Remind students to show their work and explain their reasoning, especially for free response questions.
- Encourage students to use real-world examples to illustrate their understanding of percent increase and decrease.